

WHEN SPACECRAFT THINK FOR THEMSELVES:

The Rise Of Self-Healing, Sensor-Fused Navigation

As missions stretch farther from Earth, spacecraft must think on their own, diagnosing faults and fusing sensors to stay accurate and resilient. The smarter the electronics and algorithms become, the farther we can safely go.



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Space missions are multiplying across smallsat constellations, ambitious lunar and cis-lunar programmes, and deep-space explorers travelling farther into the solar system. In this environment, robust autonomy has moved from ‘useful’ to ‘essential.’ Navigation serves as the spacecraft’s nervous system, and when operations continue with minimal ground intervention, resilience depends on two core capabilities: autonomous fault diagnosis and adaptive multi-source sensor fusion.

Together, these capabilities enable navigation systems that detect anomalies, adapt to degraded or failing sensors, and sustain precise state estimation amid the noise, extremes, and unpredictability of space. The focus spans the electronics backbone, the

algorithms that drive adaptation and diagnosis, mission case studies that validate performance in orbit, the recent advances that make such systems practical, and the developments likely to shape the next decade of resilient spacecraft navigation.

The foundation of modern spacecraft navigation electronics

Contemporary spacecraft navigation integrates diverse electronic systems, including inertial measurement units (IMUs), star trackers, radiometric tracking, and emerging modalities such as X-ray pulsar navigation, optical terrain correlation, and inter-satellite ranging. Supporting electronics have progressed from discrete radiation-hardened components to compact, power-efficient system-on-chip designs with analogue front-ends, digital